Trần Mạnh Duy

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Profile

Backend Engineer with 3 years of experience building high-throughput, event-driven Go services for correctness-sensitive fintech platforms. Experienced in owning services end to end, from re-architecting inherited codebases to diagnosing production incidents across pipelines processing millions of messages daily.

Skills

Languages: Go (primary), C++, Rust

Backend & APIs: gRPC (protobuf), Gin, net/http, REST, WebSocket

Streaming & Data: Apache Kafka (consumer groups, partitions, batching, lag tuning), Redis (sorted sets, caching, pub/sub), MongoDB, PostgreSQL

Architecture: Hexagonal / clean architecture, event-driven microservices, idempotent stream processing, state machines, structured logging

DevOps & Tooling: Docker, Git, CI/CD (GitHub Actions), Linux, k6 load testing

Testing: Table-driven unit tests, mocks/interfaces, integration tests, benchmark profiling (pprof)

Experience

Sotatek — Hanoi, Vietnam

Backend Engineer — *Jul 2023 – Present*

RaidenX — High-throughput trading data platform (multi-chain DEX aggregator)

- Owned the consumer-side data pipeline end to end (team of 8–10): Kafka consumption → PnL recompute → Redis-backed leaderboard rankings → price-alert state machine.
- Re-architected the `insight` trading-data service from an inherited single-chain TypeScript prototype into a production Go service spanning 6 blockchain networks, driving the migration largely independently.
- Diagnosed and resolved a critical Kafka consumer-lag incident (~4 M message backlog), cutting steady-state lag from millions to a few thousand.
- Redesigned ingestion from per-message (1:1) to batched processing, reducing consumer lag to the low hundreds and raising throughput ~10x.
- Ensured per-user PnL correctness across ~20–25 Kafka partitions. Grouped batched events in-memory by user/symbol before processing, preserving ordered updates despite cross-partition delivery.
- Built the PnL/profit recompute consumer on a money-critical stream with idempotent writes and deduplication under at-least-once delivery.
- Designed a per-user, preference-driven price-alert state machine that prevented duplicate alerts and powered real-time notifications.

Technology and related tools: Go, Apache Kafka, Redis, MongoDB, PostgreSQL, gRPC, Docker

Centralized Exchange Platform (Binance-style order-matching system)

- Implemented a referral-reward Kafka consumer: fetched trade data via gRPC, computed volume-based commissions, and persisted rewards with upsert-with-ignore semantics for idempotent crediting.
- Engineered a notification pipeline (Kafka → Firebase) with time-window batching to suppress duplicate pushes, backed by a Redis-cached preference store refreshed via pub/sub.

Technology and related tools: Go, Kafka, Redis, MongoDB, PostgreSQL, gRPC, Firebase

Kotoba Press — Japanese-Learning Platform

- Designed and built a Japanese-learning application on a **hexagonal (ports & adapters) architecture** in Go/Gin, with vocabulary review, grammar drills, and spaced-repetition scheduling.
- Engineered a custom full-text search engine in C++ (15K docs/sec indexing, p50 = 0.04ms at 30K qps), integrated via **gRPC (protobuf)**.
- Load-tested with **k6**: **19K req/s** on health endpoints, **4.4K req/s** on vocabulary queries (p99 < 32ms), validated linear scaling to 800 concurrent users.

Technology and related tools: Go, Gin, gRPC, MongoDB, C++, k6, Docker

Education

University of Engineering and Technology, VNU (UET-VNU) — Hanoi, Vietnam

B.Sc. Information Technology